

# JULIA WEPPLER

weppler.j@northeastern.edu | <https://julia-weppler-1.github.io/my-portfolio/>

## EDUCATION

### Northeastern University, Khoury College of Computer Sciences

Boston, MA

*Master's in Computer Science, 4.0/4.0*

September 2025 – May 2027

Northeastern Graduate Research Apprenticeship Scholarship Recipient

Spring 2026

### Boston College, Morrissey College of Arts and Sciences

Chestnut Hill, MA

*Bachelor of Science in Computer Science, Minor in Environmental Studies, 3.78/4.0*

May 2025

**Relevant Courses:** Information Visualization, Foundations of AI, Web App Development

## WORK EXPERIENCE

### Liberty Mutual

Boston, MA

*Catastrophe Modeling & GIS Research Co-op*

July 2026 – December 2026

- Built a heat hazard layer in ArcGIS for the Risk Financing team, applying NOAA heat index methodology to Copernicus Climate Data Store projections to estimate annual days above [X]°F by ZIP code
- Used interpolation techniques to create a global hazard layer of peak-ground acceleration (PGA) data from the Global Earthquake Model Foundation

### Stockholm Environment Institute

Somerville, MA

*Climate Policy Program Intern*

February 2025 – May 2026

- Lead software development for data workflows supporting renewable energy policy and industry transitions research
- Build and maintain pipelines for structured research data (cleaning, validation, automated reporting)
- Advised UNDP's Derisking Renewable Energy Investments (DREI) initiative on technology integration for trainings and workshops with community leaders and program representatives
- Researched One Health-aligned food systems to identify pathways for scaling bivalve and seaweed aquaculture; co-authored a comment submitted to *Nature Food*

### Optum (United Health Group)

Basking Ridge, NJ

*Software Engineering Intern*

June 2023 – August 2024 (Seasonal)

- Refactored Databricks notebooks into modular JARs, cutting client delivery time 60%
- Built a site feature using React for the cybersecurity department to integrate LLMs for easier graph database traversal
- Recognized as a **Standout Summer Intern**

## PROJECTS

### Graduate Research Apprenticeship Scholar, Dr Michelle Borkin | Northeastern University

January 2026

- Collected and curated statewide K-12 STEM standards and policy documents from public databases; maintained organized metadata to support cross-state comparisons
- Conducted qualitative and inductive thematic analysis to identify patterns and variation in visualization/data literacy expectations across standards

### Visualizing Physiologic Synchrony (TRACE) | Northeastern University

December 2025

- Built a Plotly Dash dashboard for clinicians interpreting parent-child physiologic synchrony, engineering synchronized video-visualization playback where timestamps drove heatmap cursors and point-in-time views
- Prototyped designs in Figma and ran iterative feedback studies with clinicians; conducted interview analysis and task abstraction in Miro to define research needs and product requirements

### F3-EPO | Stockholm Environment Institute

June 2025

- Built an interactive modeling dashboard using **React + D3** for policy analysts to explore equity-based fossil fuel phaseout scenarios
- Designed interaction patterns (linked views, overlays, tooltips) to surface trade-offs and improve decision clarity

### Data Visualization Study under Dr. Nam Wook Kim

May 2024

- Created interactive, reusable charting components to build a dashboard for analyzing categorical data from surveys
- Built a web-based sample dashboard using React and D3 for styling, connecting components, and animation

## SKILLS

**Data Management:** SQL, Excel, MongoDB, schema design, documentation, data cleaning

**Programming:** Python, Java, JavaScript, Git

**Design & Visualization:** Figma, Miro, Esri ArcGIS, Tableau, D3, Plotly Dash, Altair

**Research & Communication:** task analysis, stakeholder communication, thematic analysis, user studies